250 Homework #3 (Fall 2026)

Toby Hansen *

This assignment is due at 11:59 p.m. on Friday 23 October 2026, to be submitted on Gradescope. Late assignments (with penalty, unless excused) will be accepted until 11:59 p.m. on Saturday 24 October 2026, and not after. Please see the syllabus for more details on homework grading and late penalties.

When submitting to Gradescope, please do not forget to assign pages to problems.

All problems, with the exception of problem 2, originally come from the textbook. Some of them have been modified. In case of doubt, the version printed here is the one you should follow.

The instructions for this homework are the same as for HW1 and HW2. Please read them there. Also, do not forget to fill out the questionnaire describing collaboration and use of LLMs that appears as a separate submission on Gradescope.

See the Canvas Homework web page for detailed instructions on how to format solutions. Solutions that do not follow those formatting instructions will be returned for rewriting and resubmission. In later assignments there will be points deducted for such formatting errors.

*Collaborated with Nobody.

Q1. Modified from P3.1.1 [5 pts]

The primes have a strange distribution. See, e.g., `https://en.wikipedia.org/wiki/List_of_prime_numbers` for a list of the first 1000 primes. It shows that, the primes are scattered. Sometimes two consecutive primes only differ by two, e.g., 7589 and 7591. Sometimes they differ by quite a lot and you get a long run of composite numbers. E.g., 7759 is prime and the next prime after it is 7789, so the run of 29 numbers starting with 7760 are all composite.

For this problem find 75 consecutive naturals that are all composite. (**Hint**: The **factorial** of a natural k, written $k!$, is the product $1 \cdot 2 \cdot ... \cdot k$. You can tell a lot about which numbers divide $k!$.)

Note: To solve this problem you only need to write down the first number in the series and state that the next 74 numbers are all composite, explaining why you know this. The first item can be written using factorial notation and summation.

Solution:

Any number k's factorial can be divided evenly by all positive numbers less than or equal to k, and adding any number n to the factorial that is between 2 and k will result in a number that is still evenly divisble by n, which means it is composite. Using this logic, we can easily find k consecutive composite naturals where k is greater than 1 using $(k+1)! + n$ where n is between 2 and $k+1$. $k+1$ must be used instead of just k because it isn't a guarantee that $k! + 1$ or $k! + k + 1$ are composite.

With this in mind, to find 75 consecutive composite naturals, we just have to find:

$$\{76! + 2, 76! + 3, 76! + 4, \ldots, 76! + 76\}$$

Q2. New Problem from Section 3.5 [10 pts]

(a) Use the extended gcd algorithm to find integers a and b satisfying $a \cdot 137 + b \cdot 111 = 1$. Show all steps of the algorithm.
For full credit you must use the exact technique we taught in class, which is designed to catch errors early. Use of a different technique will lead to points being deducted, even for a correct final result.

Note: You should check the solution of (a) before proceeding to (b) and (c). Errors in (b) and (c) resulting from finding an incorrect a, b pair in part (a) will not receive partial credit.

(b) Find a multiplicative inverse of 137 modulo 111 and a multiplicative inverse of 111 modulo 137. The inverses you derive should be the smallest natural numbers that are inverses.

(c) Find the smallest natural solution to the congruences $x \equiv 2 \pmod{137}$ and $x \equiv 3 \pmod{111}$. Show your work.

Solution:

(a) $137 \mod 111 = 26$
$111 \mod 26 = 7$
$26 \mod 7 = 5$
$7 \mod 5 = 2$
$5 \mod 2 = 1$
$2 \mod 1 = 0$

$26 = 137(1) + 111(-1)$
$7 = 111(1) + 26(-4) = 111(1) + 137(-4 * 1) + 111(-4 * -1) = 137(-4) + 111(5)$
$5 = 26(1)+7(-3) = 137(1)+111(-1)+137(-3*-4)+111(-3*5) = 137(13)+111(-16)$
$2 = 7(1)+5(-1) = 137(-4)+111(5)+137(-1*13)+111(-1*-16) = 137(-17)+111(21)$
$1 = 5(1) + 2(-2) = 137(13) + 111(-16) + 137(-2 * -17) + 111(-2 * 21) =$
$137(47) + 111(-58)$
Thus, $a = 47$, and $b = -58$

(b) The inverse for $137 \mod 111$ is already given by a (47). Since b (-58) is negative, we can add 137 to get the *natural* inverse of $111 \mod 137$, which is 79.

(c) By Chinese Remainder Theorem, $c = 137(47)(3) + 111(-58)(2) = 19317 - 12876 = 6441$, removes the need to brute force a solution:
$137(0) + 2 = 2$ and $111(0) + 3 = 3$
$137(1) + 2 = 139$ and $111(1) + 3 = 114$
...
$137(47) + 2 = 6441$ and $111(58) + 3 = 6441$

Thus, $x \equiv 6441 \mod (137 * 111)$, so, $x \equiv 6441 \mod (15207)$
Since 6441 is less than 15207, 6441 must be the smallest natural solution.

Q3. Modified from P3.3.4 [10 pts]

We have defined the **factorial** $n!$ of a natural n to be the product of all the naturals from 1 through n, with 0! being defined as 1. Let p be an odd prime number.

(a) Prove that the only solutions to $x^2 \equiv 1 \pmod{p}$ satisfying $0 < x < p$ are $x = 1$ and $x = p - 1$.

Hint: *Note that* $x^2 \equiv 1 \pmod{p}$ *is equivalent to* $(x^2 - 1) \equiv 0 \pmod{p}$ *and that* $x^2 - 1 = (x - 1)(x + 1)$.

(b) Prove that $(p - 1)!$ is congruent to -1 modulo p.

Hint: *Use part (a) and also pair as many numbers as you can with their multiplicative inverses.*

Examples: $4! = 24$, $6! = 720$ *and* $10! = 3628800$ *while* $24 \equiv -1 \pmod{5}$, $720 \equiv -1 \pmod{7}$, *and* $3628800 \equiv -1 \pmod{11}$

Solution:

(a) Since $(x - 1)(x + 1) \equiv 0 \mod p$, $x - 1$ and $x + 1$ must be multiples of p.
First, since $0 < x < p$, $x - 1$ must be greater than or equal to 0, but less than $p - 1$; Since p is greater than 2 (on account of being an odd prime), the only multiple of p in that range is technically 0, $x - 1 = 0$ so $x = 1$.
Second, since $0 < x < p$, $x + 1$ must be greater than 1, but less than $p + 1$; Since p is greater than 2 (on account of being an odd prime), the only multiple of p in that range is p itself, $x + 1 = p$ so $x = p - 1$.

(b) From part (a), the only naturals that are their own multiplicative inverses $\mod p$ (between zero and p) are 1 and $p - 1$, so every natural from $\{2, 3, \ldots, p - 2\}$ can be paired with another natural in the set so that their product $\mod p = 1$. For example, if $p = 7$, then 2 can be paired with 4 since $8 \mod 7 = 1$, and 3 can be paired with 5 since $15 \mod 7 = 1$.
Since each pair's product $\mod p = 1$, the product of all of them $\mod p$ must also be equal to 1; but that's just (p-2)!, so multiply it by p-1 to get (p-1)!, and since p-1 is its own multiplicative inverse, it can be paired with itself (in the earlier example $p - 1 = 6$, $36 \mod 7 = 1$).
In practice, $6! = 1 * 2 * 3 * 4 * 5 * 6$, but since $(2 * 4)$ and $3 * 5 \mod p = 1$, only 1 and 6 remain when the modulo is taken: $6! \equiv 1 * 1 * 1 * 6 \mod 7$, which is the same as saying $6! \equiv 6 \equiv -1 \mod 7$. 7 was an arbitrary odd prime, but the example can generalized to all odd primes based on the logic of the set between 1 and $p - 1$ being uniquely pairable.

Q4. P3.5.10 Modified [15 pts]

Lecture 18 will discuss how to use *induction* to prove facts about a particular change-making problem. In this question you only need to use number-theoretic facts and not induction.

Let m and n with $m < n$ be two positive naturals. We say that natural x is (m, n)-*good* if x can be expressed as a linear combination $a \cdot m + b \cdot n$ where a and b are naturals, i.e., non-negative integers.

In parts (c) and (d), $[x]$ denotes the equivalence class of x modulo m.

Parts (a) and (b) introduce you to the setup. Parts (c)-(f) are proving the Theorem that "There are an infinite number of (m, n)-*bad* naturals if and only if m and n are not relatively prime."

a) Let $m = 2$ and $n = 3$. Prove that all naturals except 1 are $(2, 3)$-*good* and 1 is $(2, 3)$-*bad*.

b) Determine all naturals which are $(3, 5)$-*good* and all which are $(3, 5)$-*bad*. Prove your statement.

c) Let m, n be relatively prime and $i \in \{0, 1, 2, \ldots, m - 1\}$. Prove that there for every such i there exists natural j_i such that $j_i \cdot n \equiv i \pmod{m}$.
Equivalently prove that there exists a natural j satisfying $[j_i \cdot n] = [i]$.

d) Suppose that x is (m, n)-*good*. Prove that if $y \geq x$ and $y \in [x]$, then y is (m, n)-*good*.

e) Let m, n be relatively prime. Prove that only a finite number of naturals are (m, n)-*bad*.
Equivalently, prove that there exists natural N, such that if $x \geq N$, x is (m, n)-*good*.

f) Suppose m, n are NOT relatively prime.
Prove that the set of (m, n)-*bad* naturals is infinite.

Hints and comments:

1. *0 is always (m, n)-good for all m, n.*
2. *For (c), you may either prove from scratch or use any fact that we have proven in class.*
3. *For (e), use parts (c) and (d). If you get stuck, you might want to look ahead to the example shown at the end of Lecture 18 to see how this can be helpful.*
4. *For (f), construct an infinite sequence, all of whose elements are (m, n)-bad.*

Solution:

(a) Since (m, n) is good only when x is a combination of multiples of m and n, and EVERY natural except for 1 can be represented by a combination of multiples of m and n (even zero), all naturals except 1 are $(2, 3)$-*good*. To cover some base cases, $2 = 1(2) + 0(3)$, $3 = 0(2) + 1(3)$, $4 = 2(2) + 0(3)$, $5 = 1(2) + 1(3)$, etc; for any natural n (other than 1), if it's even it can be represented as a multiple of just 2 like $k(2) + 0(3)$ where $k = n/2$, if

it's odd it can be represented using 2 to get to $n-3$ then add a single 3 like $j(2)+1(3)$ where $j=(n-3)/2$. Any number lower than 2 can't be represented however, with the exception of zero, since $0=0(2)+0(3)$; since 1 the only number lower than 2 other than zero, it can't be represented my m and n and is therefore (m,n)-*bad*.

(b) To cover some base cases, $0=0(3)+0(5)$, $3=1(3)+0(5)$, $5=0(3)+1(5)$, $6=2(3)+0(5)$, so $\{0,3,5,6\}$ are $(3,5)$-*good*, and $\{1,2,4,7\}$ are $(3,5)$-*bad* since they can't be represented by $3a+5b$. Now, by adding 3 or 5 to any confirmed $(3,5)$-*good* natural, it can be represented by just increasing a or b by 1, and since EVERY natural greater than 7 can be acquired by adding multiples of 3 and five to $\{0,3,5,6\}$, every natural greater than 7 is $(3,5)$-*good*, therefore $\{1,2,4,7\}$ are the only $(3,5)$-*bad* naturals.

(c) Since m and n are relatively prime, $\gcd(m,n)=1$, so by extended gcd, a and b must exist such that $am+bn=1$, so $bn\equiv 1 \mod m$, meaning b is a multiplicative inverse of $n \mod m$. Multiplying both sides by i gives $ibn\equiv i \mod m$. We can just make $j_i=ib$, which proves that for every such i there exists natural j_i such that $j_i\cdot n\equiv i \pmod m$. Since j_i is a multiple of i (ib, to be precise), we can equivalently find a version with an even divisor instead, which satisfies $[j_i\cdot n]=[i]$.

(d) Since x is (m,n)-*good*, there exists naturals a and b such that $x=am+bn$. Since $y\geq x$ and $y\equiv x \mod m$, subtracting x from both sides gives $y-x\equiv 0 \mod m$ so $y-x$ is evenly divisible by m. Therefore, $y-x=km$ where k is some natural. Substituting $x=am+bn$ into $y=x+km$ gives $y=am+bn+km=m(a+k)+bn$. Since $a+k$ and b are both naturals, y is also (m,n)-*good*.

(e) Since m and n are relatively prime, from part (c) we know that for every remainder i there exists a natural j_i such that $j_in\equiv i \mod m$. Each j_in is (m,n)-*good* because $j_in=0(m)+j_i(n)$. Since there are only m possible remainders let N be the largest. Now, for any natural $y\geq N$, let i be the remainder of $y \mod m$. We know $y\equiv j_in \pmod m$, and since $y\geq N\geq j_in$, part (d) says y is also (m,n)-*good*. Therefore, EVERY natural greater than or equal to N is (m,n)-good, meaning there's a finite amount of naturals between 0 and N that could be (m,n)-*bad*.

(f) Since m and n are not relatively prime, $d=\gcd(m,n)>1$. Since d divides both m and n, it must also divide every combination of $am+bn$. Therefore, every (m,n)-*good* natural must be divisible by d. However, there are infinitely many natural numbers that are not divisible by d ($fd+1$ where f is any natural). None of these numbers can be written as $am+bn$, so they are all (m,n)-*bad*. Therefore, there are infinitely many (m,n)-*bad* naturals when m and n are not relatively prime.

Q5. P3.6.4 [10 pts]

In this problem we will work with strings over the alphabet $\{a, b\}$, and let our "multiplication" be string concatenation.

We say that one string u "divides" another string v if $\exists x : \exists y : xuy = v$, or equivalently, if u is a substring of v. This is denoted by $D(u, v)$.

As with the naturals, we redefine the word "prime" for strings so that $P(w)$ means "$w \neq \lambda$ and if $D(x, w)$ is true, then either $x = \lambda$ or $x = w$"

(a) What strings are "prime" using this definition?

(b) Explain why any string factors uniquely as a product (concatenation) of these "primes".

(c) Prove a version of the Atomicity Lemma for these "primes".

(d) Show that any nonempty string that is *not* one of your "primes"" fails to satisfy the Atomicity Lemma, even in the simple version.

Notes:

(i) You can prove (c) by following the structure of the proof of the original atomicity lemma in the book and first proving a simple version and then a complete version.

(ii) The proof of the simple version should be easier than the proof for naturals. You can write it in words. You may assume that given a character 'c' and a string 's', you are always able to tell whether 'c' is in 's' or not.

Solution:

(a) All possible strings are $\{\lambda, a, b, aa, ab, ba, bb, aaa, etc...\}$ just like binary. The only prime strings using the given definition are just p[lain a and b. Since there are an infinite number of combinations that can all "divide" each other, only the building blocks themselves are prime.

(b) Since a and b are the only prime strings, every nonempty string can be factored into a sequence of these primes, and the order matters. For example, $abba = a * b * b * a$, NOT $a * a * b * b$; even though it has the same amount of the prime characters, the order would yield a different unique string. Therefore, every nonempty string factors uniquely. Also, the nonempty string is just the lack of either of the primes and is still unique itself.

(c) If a prime divides a product, then it must divide at least one of the factors. In this case, the prime strings are only a and b. Suppose one of the prime strings p divides the concatenation uv. Since p consists of exactly one character, it cannot be a substring between u and v. Therefore, that character must appear entirely independently within u, v, or both, meaning $D(p, u)$ or $D(p, v)$ or both.

(d) For any nonempty non-prime string w, it must be at least 2 characters long. We can split w into two nonempty strings u and v, where u is the first character and v is the remaining characters, so $w = uv$. $D(w, uv)$ is true because $uv = w$. However, $D(w, u)$ and $D(w, v)$ are both false because w is longer than both, so it can't be a substring of either.

Q6. P4.1.6 [10 pts]

(uses Java) Give a recursive definition of and a recursive static method for the natural subtraction function, with pseudo-Java header

`natural minus (natural` x`, natural` y`).`

On input x and y this function returns $x-y$ if this is a natural (i.e., if $x \geq y$) and 0 otherwise.

Solution:

```
// This function returns the difference between between two naturals, but only
// when x is greater than or equal to y. Otherwise, it treats them as equal.
// This is accomplished by repeatedly subtracting 1 from both both naturals
// until y reaches zero, at which point x is equal to the raw difference.
natural minus(natural x, natural y) {
    if (y == 0) return x;
    if (x == 0) return 0;
    return minus(x - 1, y - 1);
}
```

If $y = 0$, then the result is x, since nothing remains to be subtracted. If $x = 0$ while $y > 0$, then the result is 0, since the subtraction would otherwise be negative.

Q7 Modified 4.3.1 [10 pts]

Define $S_n = \sum_{i=0}^{n} i^3$, i.e., as the sum of the first i cubes . Prove by induction that, for all naturals n,

$$S(n) = \frac{n^2(n+1)^2}{4}. \tag{1}$$

Your proof should be split into three parts

(i) State your inductive hypothesis $P(n)$.
(This must explicitly state what set n comes from.)

(ii) State and justify the correctness of the base case.

(iii) Prove the correctness of the induction step.
This must explicitly state exactly in which step(s) the proof is using the inductive hypothesis.
Also, every line of your proof should follow easily from the previous line. It's your job to provide justification of each step from the previous one, not the reader's.

Solution:

(i) Assume that the statement is true for some arbitrary natural k, meaning $\sum_{i=0}^{k} i^3 = \frac{k^2(k+1)^2}{4}$.

(ii) For the base case, let $n = 0$. The left side is $\sum_{i=0}^{0} i^3 = 0^3 = 0$, and the right side is $\frac{0^2(0+1)^2}{4} = 0$. Since both sides are equal, the statement holds for $n = 0$.

(iii) For $k + 1$:

$$\begin{aligned}
\sum_{i=0}^{k+1} i^3 &= \sum_{i=0}^{k} i^3 + (k+1)^3 && \text{(expand the summation)} \\
&= \frac{k^2(k+1)^2}{4} + (k+1)^3 && \text{(inductive hypothesis)} \\
&= \frac{k^2(k+1)^2 + 4(k+1)^3}{4} && \text{(multiply right term by 4/4, combine)} \\
&= \frac{(k+1)^2(k^2 + 4(k+1))}{4} && \text{(take out like terms)} \\
&= \frac{(k+1)^2(k^2 + 4k + 4)}{4} && \text{(distribute the 4)} \\
&= \frac{(k+1)^2(k+2)^2}{4}. && \text{(factor)}
\end{aligned}$$

Q8 Modified 4.3.3 [10 pts]

Let the finite sequence $a_0, a_1, \ldots, a_n$ be defined by the rule $a_i = a_0 \cdot r^i$, where r is some number not equal to 1. Prove by induction on n that

$$\sum_{i=0}^{n} a_i = \frac{a_0 - a_{n+1}}{1 - r},$$

where a_{n+1} is defined by the same rule even though it is not part of the sequence.

Note: This is just the formula for the sum of a geometric sequence. But, you cannot use that fact. The problem is asking you to prove this directly from scratch using simple induction.

Your proof should be split into three parts

(i) State your inductive hypothesis $P(n)$.

(ii) State and justify the correctness of the base case.

(iii) Prove the correctness of the induction step.
This must explicitly state exactly in which step(s) the proof is using the inductive hypothesis.
Also, every line of your proof should follow easily from the previous line. It's your job to provide justification of each step from the previous one, not the reader's.

Solution:

(i) Assume that the statement is true for some arbitrary natural k, meaning $\sum_{i=0}^{k} a_i = \frac{a_0 - a_{k+1}}{1-r}$, where $a_i = a_0 r^i$ and $r \neq 1$.

(ii) For the base case, let $n = 0$. The left side is $\sum_{i=0}^{0} a_i = a_0$, and the right side is $\frac{a_0 - a_1}{1-r} = \frac{a_0 - a_0 r}{1-r} = \frac{a_0(1-r)}{1-r} = a_0$. Since both sides are equal, the statement holds for $n = 0$.

(iii) For $k + 1$:

$$\begin{aligned}
\sum_{i=0}^{k+1} a_i &= \sum_{i=0}^{k} a_i + a_{k+1} && \text{(expand the summation)} \\
&= \frac{a_0 - a_{k+1}}{1 - r} + a_{k+1} && \text{(inductive hypothesis)} \\
&= \frac{a_0 - a_{k+1} + a_{k+1}(1 - r)}{1 - r} && \text{(combine using common denominator)} \\
&= \frac{a_0 - a_{k+1} + a_{k+1} - a_{k+1} r}{1 - r} && \text{(distribute)} \\
&= \frac{a_0 - a_{k+1} r}{1 - r} && \text{(cancel like terms)} \\
&= \frac{a_0 - a_{k+2}}{1 - r}. && \text{(since } a_{k+2} = a_{k+1} r\text{)}
\end{aligned}$$

EC: P3.4.6 [10 pts]

A **Fermat number** is a natural of the form $F_i = 2^{2^i} + 1$, where i is any natural. In 1730 Goldbach used Fermat numbers to give an alternate proof that there are infinitely many primes.

(a) List the Fermat numbers F_0, F_1, F_2, F_3, and F_4.

(b) Prove that for any n, the product $F_0 \cdot F_1 \cdot \ldots \cdot F_n$ is equal to $F_{n+1} - 2$.

Hint. we recommend proving this by induction. If you do prove it via induction, don't forget to split it into three parts, as in problems 7 and 8.

(c) Argue that no two different Fermat numbers can share a prime factor. Since there are infinitely many Fermat numbers, there must thus be infinitely many primes.

Solution:

(a) The The first five Fermat numbers are:

$$F_0 = 2^{2^0} + 1 = 3$$
$$F_1 = 2^{2^1} + 1 = 5$$
$$F_2 = 2^{2^2} + 1 = 17$$
$$F_3 = 2^{2^3} + 1 = 257$$
$$F_4 = 2^{2^4} + 1 = 65537.$$

(b)

(c)